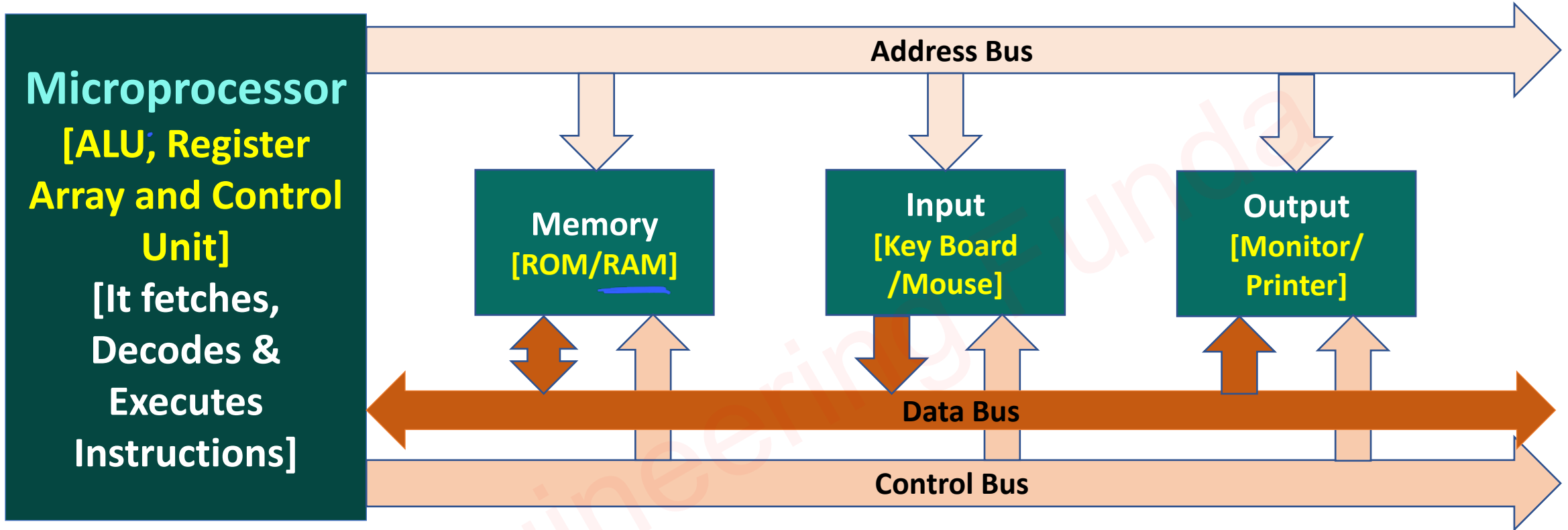


System Buses in Computer Organization



❖ Address Bus

- ❑ It is generated by μP .
- ❑ These buses are used to Identify the address of Memory and IO peripherals.
- ❑ For 16 lines of Address bus, we can interface $2^{16} = 64\text{KB}$ memory.

❖ Control Bus

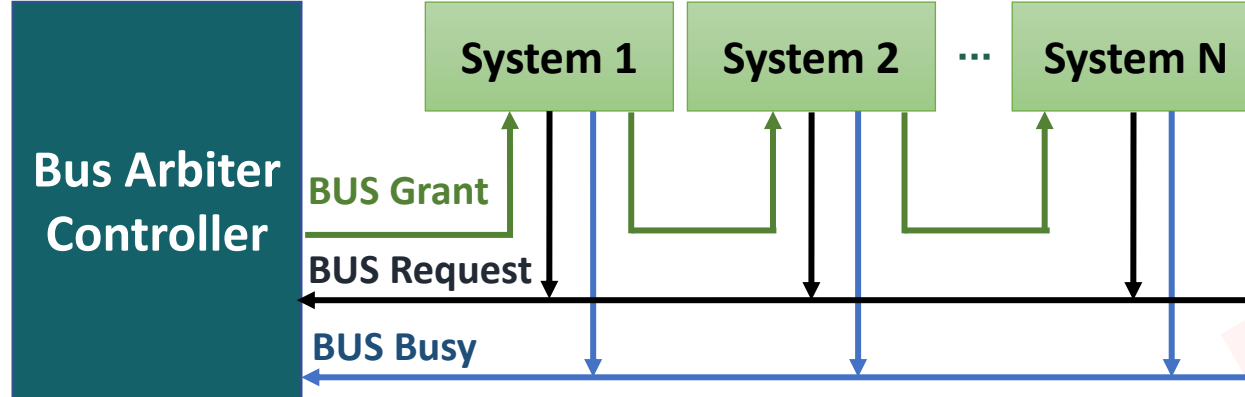
- ❑ It is generated by μP .
- ❑ Control Bus explains operations.
- ❑ Memory Read, Memory write, IO Read & IO write are the basic control operations on system bus.

❖ Data Bus

- ❑ It is bidirectional buses.
- ❑ Direction of buses is based on read and write operation.
- ❑ Data bus explains how much data can be accessed by processor in single machine cycle.

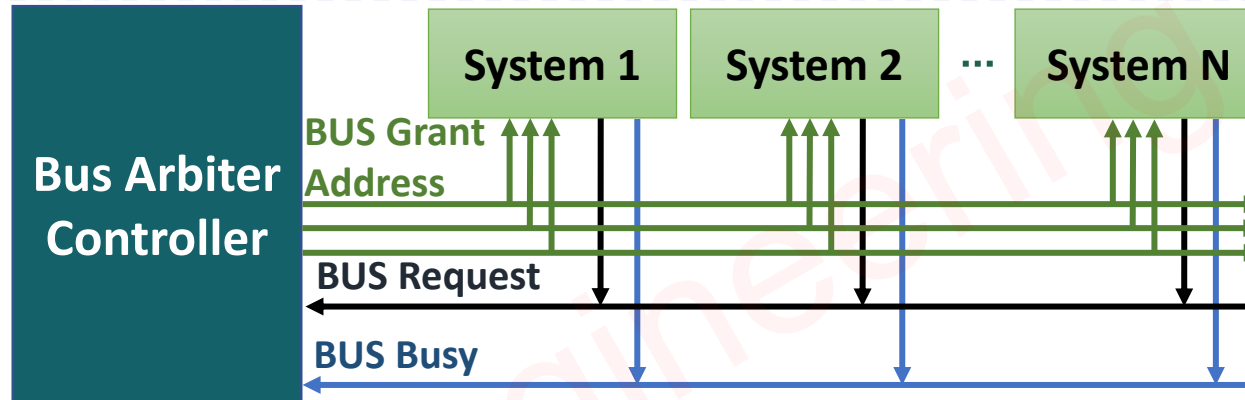
Daisy Chain, Polling & Independent Request

Daisy Chain Method



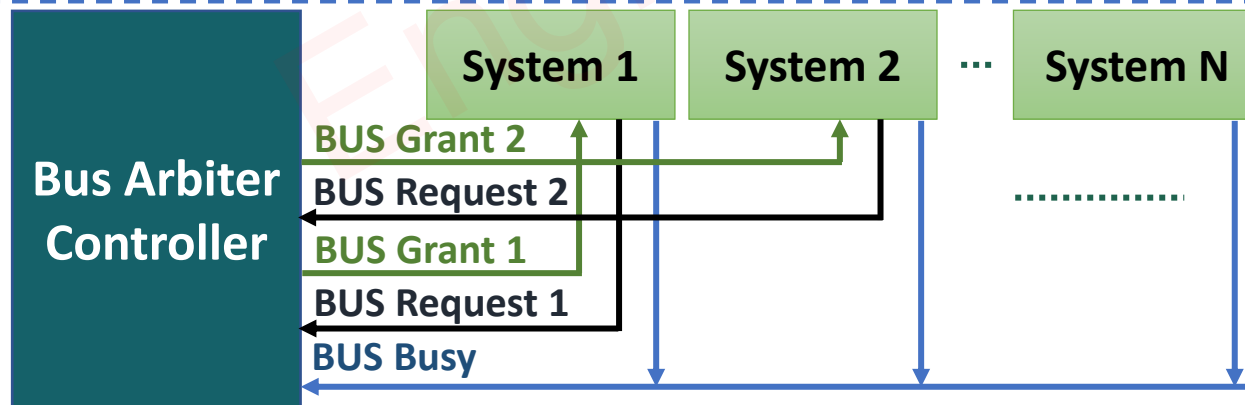
- ☐ All Bus Masters have same line for Bus request.
- ☐ If Bus Busy line is inactive then Bus controller gives the bus grant signal.
- ☐ Bus grant signal transmitted Serially through all the Systems or Bus Masters.
- ☐ If Bus Master requires system Bus then it will activate the bus busy signal and takes control of system bus.

Polling Method



- ☐ All Bus Masters have same line for Bus request.
- ☐ If Bus Busy line is inactive then Bus controller gives the bus grant Address based on Bus request.
- ☐ Bus Controller Polls the System in well define order as per priority.
- ☐ Once System receives it's own address, it will active the bus busy signal and takes control of system bus.

Independent Request Method



- ☐ All Bus Masters have individual bus request lines.
- ☐ So, Controller knows which System is asking for Bus request.
- ☐ Priority of all systems is predefined.
- ☐ So based Bus availability and priority Bus grant is given to systems.
- ☐ Controller consists encoder and decoder logic for priorities.

Comparison of Daisy Chain, Polling & Independent Request

Parameters	Daisy Chain	Polling	Independent Request
Design	Simplest	Little Complex	Complex
Reliability	Not Reliable if any system fails.	Fully Reliable	Fully Reliable
Performance	Weak	Better then Daisy Chain	Excellent
Upgradation	Easy	Reconfiguration requires	Complexity Increases
Cost	Minimum	Less	High
Lines for 8 systems	Bus Grant = 1 Bus Request = 1 Bus Busy =1	Bus Grant = 3 Bus Request = 1 Bus Busy =1	Bus Grant = 8 Bus Request = 8 Bus Busy =1

